

CS 505 Module Two Activity Template

Complete this template by replacing the bracketed text with the relevant information.

Set Theory Calculations

1. Determine the number of elements in the following sets:
 - A. $\{\{a, b, \{a, b\}, \{a, c\}\}\}$
 - i. This set has 1 element: $\{a, b, \{a, b\}, \{a, c\}\}$
 - B. $\{\{\}, \{\{\}\}, \{\}$
 - i. This question has 2 sets:
 1. 2 Elements: $\{\}$ & $\{\{\}\}$
 2. 0 Elements: Empty set
 - C. $\{a, b, \{\}, \{\{\{\}\}\}\}$
 - i. This set has 4 elements: $a, b, \{\}, \{\{\{\}\}\}$
2. List the members of the following sets:
 - A. $\{x|x \text{ is a positive integer less than } 10\}$
 - i. $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
 - B. $\{x|x \text{ is a real number such that } x^2 = 4\}$
 - i. $\{-2, 2\}$
 - C. $\{x|x \text{ is the square of an integer and } x < 200\}$
 - i. $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196\}$
3. Consider the following sets and determine whether each of the following statements is true or false:
 $A = \{2, 3, 5, 9, 10\}$
 $B = \{3, 5, 8\}$
 $C = \{2, 6\}$
 $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 - A. $8 \in B$
 - i. True, 8 is an element of set B.
 - B. $B \subset A$
 - i. False, B is not a subset of A. 3 and 5 are elements in A, but 8 is not.
 - C. $C \subset A$
 - i. False, C is not a subset of A. 2 is an element of A, but 6 is not.
 - D. $2 \in C$
 - i. True, 2 is an element of C
 - E. $C \subset B$
 - i. False, C is not a subset of B. None of the elements in C exist in B.
 - F. $C \subset U$
 - i. True, C is a subset of U. All elements in C exist in U.

Set Theory Rationale

4. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 6\}$, and $C = \{2, 4\}$. Answer each of the following questions:
 - A. Is $B \subseteq A$?

- i. B is not a subset of A because not all elements of B are contained in A. While 1 and 2 appear in both sets, element 6 is in B but not in A. A subset requires every element of one set to be included in the other, so this condition is not satisfied.
- B. Is $C \subseteq A$?
 - i. C is a subset of A because every element in C (2 and 4) is also contained in A. This satisfies the definition of a subset, where all elements of one set must exist within another set.
- C. Is $C \subseteq C$?
 - i. C is a subset of itself because every set is always a subset of itself. This is a fundamental property of sets, as all elements in C are trivially contained within C.
- D. Is C a proper subset of B?
 - i. C is not a proper subset of B because not all elements of C are contained in B. Although a proper subset must be entirely contained within another set and not equal to it, the element 4 is in C but not in B, so the condition fails.
- 5. Explain whether each of the following is a well-defined set:
 - A. The collection of cardinal directions.
 - i. This is a well-defined set because the elements are clearly and universally agreed upon. The cardinal directions—North, South, East, and West—are objective and unambiguous, meaning there is no confusion about whether an element belongs in the set. Therefore, the set can be written as {North, South, East, West}.
 - B. A collection of interesting stamps.
 - i. This is not a well-defined set because the term “interesting” is subjective and varies among individuals. A well-defined set requires clear, objective criteria so that membership can be determined consistently.
 - C. The collection of all real numbers x for which $3x - 9 = 18$.
 - i. This is a well-defined set because the equation produces a single, unambiguous solution. Solving the equation yields $x = 9$, which clearly determines the set’s only element.
 - D. The collection of all integers x for which $5x - 10 = 3$.
 - i. This is a well-defined set because the condition is precise and produces a clear result. Solving the equation gives $x = 13/5$, which is not an integer. Therefore, no elements satisfy the condition, and the set is empty. An empty set is still considered well-defined.
 - E. The collection of all terrible basketball players.
 - i. This is not a well-defined set because the term “terrible” is subjective and lacks a consistent standard for inclusion. Without objective criteria, it is impossible to determine membership definitively.

Set Operations and Cardinality Calculations

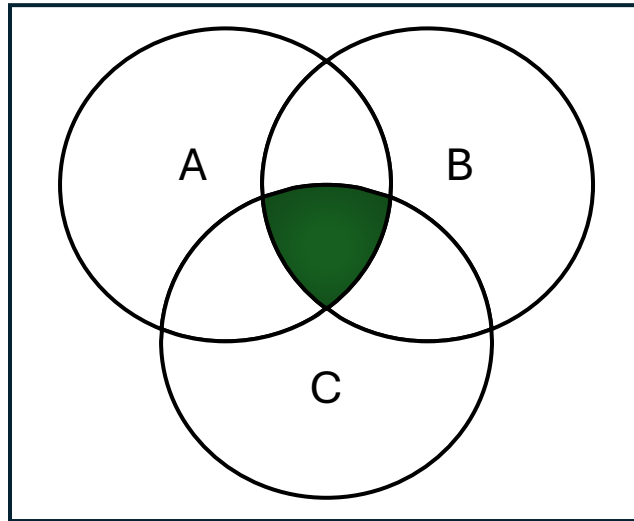
- 6. Find the number of elements in the set $\{3, 12, 15, 25\}$ and then use the formula for the cardinality of a power set to find the cardinality of $P(\{3, 12, 15, 25\})$.
 - A. $|A| = 4$

- B. $|P(A)| = 2^{|A|} = 16$
7. Find the union of two sets: $A = \{2, 5, 9, 11\}$ and $B = \{2, 5, 11, 25\}$.
A. $A \cup B = \{2, 5, 9, 11, 25\}$
8. Find the intersection of sets $P = \{5, 6, 9, 11\}$ and $Q = \{6, 11\}$.
A. $A \cap B = \{6, 11\}$
9. Find the complement of set $X = \{p, r, s\}$ where universal set $U = \{e, f, h, p, r, s\}$.
A. $X^c = \{e, f, h\}$
10. Given two sets $A = \{1, 3, 4, 7, 10\}$ and $B = \{1, 3, 8, 15\}$, find $B - A$.
A. $\{8, 15\}$
11. Consider the following sets:
 $A = \{\text{pink, green, blue}\}$
 $B = \{\text{pink, purple, orange}\}$
 $C = \{\text{pink, orange, purple, green, blue, red}\}$
 Determine each of the following:
 A. Find $A \cup B$.
 i. $\{\text{pink, green, blue, purple, orange}\}$
 B. Find $A \cap B$.
 i. $\{\text{pink}\}$
 C. Find $A^c \cap C$.
 i. $\{\text{orange, purple, red}\}$
12. Consider the following sets:
 $H = \{\text{cat, dog, tortoise, fish}\}$
 $F = \{\text{dog, cow, duck, pig, rabbit}\}$
 $W = \{\text{duck, tortoise, deer, frog, fish}\}$
 Determine each of the following:
 A. $(H \cap F) \cup W$
 i. $\{\text{dog, duck, tortoise, deer, frog, fish}\}$
 B. $H \cap (F \cup W)$
 i. $\{\text{dog, tortoise, fish}\}$
 C. $(H \cap F)^c \cap W$
 i. $\{\text{duck, tortoise, deer, frog, fish}\}$

Set Operations and Cardinality Rationale

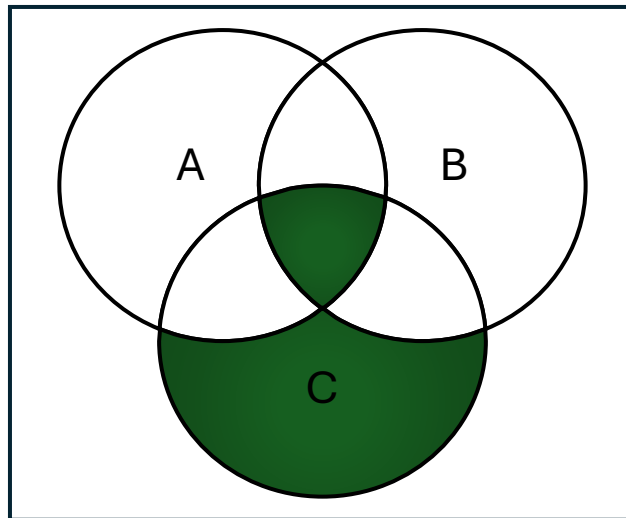
13. Let A , B , and C be subsets of a universal set U . For each of the following, draw a Venn diagram for three sets and shade the region(s) that represent the specified set.
 A. $(A \cap B) \cap C$
 i. This expression represents the elements that are common to all three sets A , B , and C . Since an intersection identifies shared elements, the result is

the region where all three sets overlap in the center of the Venn diagram.



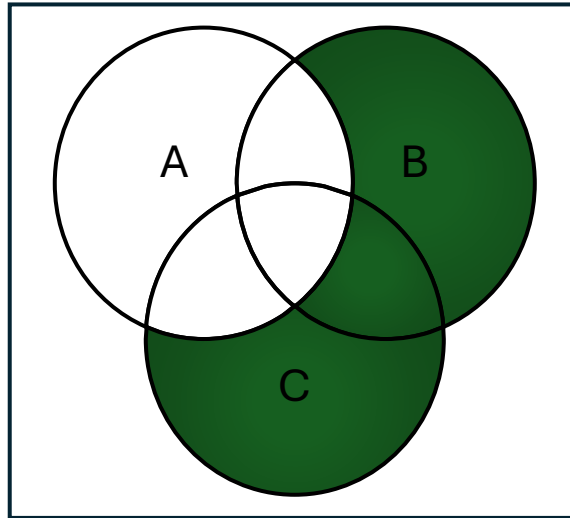
B. $(A \cap B) \cup C$

- i. This expression combines two operations. First, $A \cap B$ identifies elements shared by A and B. Then, taking the union with C includes all elements in C as well as those shared by A and B. This results in the entire region of C, along with the overlapping portion of A and B.



C. $A^c \cap (B \cup C)$

- i. This expression represents elements that are in either B or C but not in A. The union $B \cup C$ gathers all elements from both sets, and intersecting with A^c removes any elements that belong to A. The result is all elements in B and C that lie outside of A.



14. Let $A = \{1, 2, 4\}$, $B = \{1, 2\}$, $C = \{0, 5, 9\}$, and let the universal set be $U = \{0, 1, 2, \dots, 9\}$. Determine each of the following:

A. $A \cap B$

- i. The intersection of A and B includes only elements that appear in both sets. This operation filters the sets to their shared elements, resulting in $\{1, 2\}$.

B. $A \cup B$

- i. The union of A and B includes all unique elements from both sets. Duplicate elements are only listed once, resulting in a combined set of $\{1, 2, 4\}$.

C. $A \cup C$

- i. This union combines all distinct elements from sets A and C. The result includes every element that appears in either set, forming $\{0, 1, 2, 4, 5, 9\}$.

D. $A - B$

- i. The set difference $A - B$ includes elements that are in A but not in B. By removing the elements 1 and 2 (which appear in B), only 4 remains.

E. A^c

- i. The complement of A includes all elements in the universal set U that are not in A. This operation effectively removes A from U, leaving $\{0, 3, 5, 6, 7, 8, 9\}$.

F. C^c

- i. The complement of C includes all elements in the universal set that are not part of C. Removing $\{0, 5, 9\}$ from U leaves $\{1, 2, 3, 4, 6, 7, 8\}$.

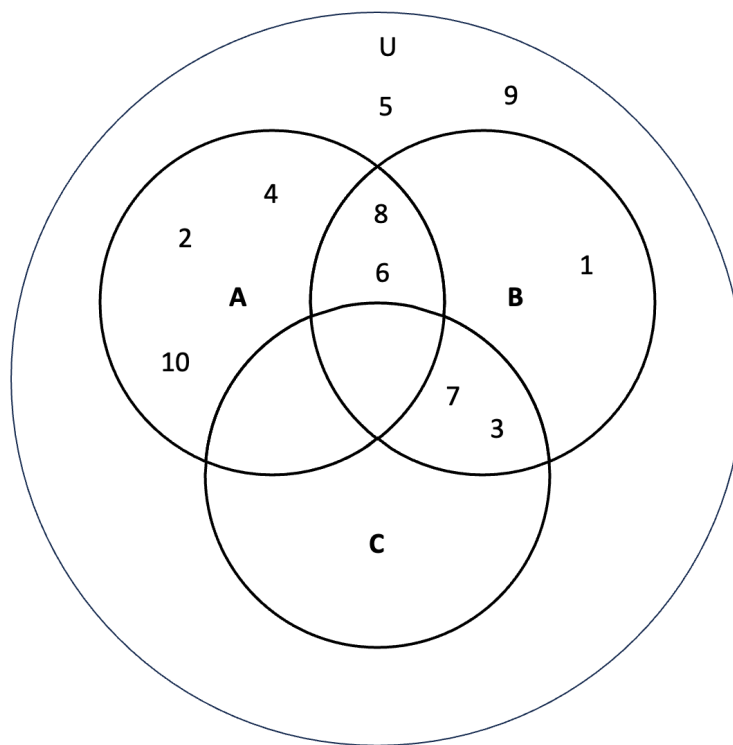
15. Illustrate the following sets in a Venn diagram, marking all the elements in the appropriate places:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 3, 6, 7, 8\}$$

$$C = \{3, 7\}$$



A. Using your Venn diagram, list the elements in each of the following sets:

- i. $A \cap B$
 $\{6, 8\}$
- ii. $A \cup C$
 $\{2, 3, 4, 6, 7, 8, 10\}$
- iii. $B \cap A^c$
 $\{1, 3, 7\}$
- iv. $B \cap C^c$
 $\{1, 6, 8\}$
- v. $C - B$
 $\{\}$